

Social Determinants of Health Impact on Patient Perception within a Home Infusion and Specialty Pharmacy Setting

¹Leslie Myers, PharmD, IgCP; ¹Michele Way, PharmD, IgCP, BCSCP; ¹Elizabeth Neal, RN, IgCN; ¹Edward O'Bryan, MD, MBA, CPE; ¹Taylor Cathcart, RN; ¹James Sheets, PharmD, IgCP; ¹Elizabeth Duruz, R.Ph, CSP, IgCP; ¹Randy Broyles, R.Ph.; ¹Timothy P. Walton, MHS, CCRP
¹CSI Pharmacy, Plano, TX

National Associate of Specialty Pharmacy (NASP) September 14 – 17, 2025, Denver, CO

BACKGROUND

- Home infusion is an alternative site of care for patients requiring the administration of medications, like immunoglobulin (Ig) and biologic therapy in a setting outside of a hospital, medical facility or outpatient infusion center. This service is supported by specialty pharmacies and home infusion companies with care provided by professionals who have obtained additional training.¹
- Administering infusions in the home setting reduces time away from everyday activities, such as work and school, offering a safe and effective alternative to traditional infusion settings, as well as offering potential cost savings.^{2,3} The range of treatments offered in the home setting continues to increase, as quality of life benefits and improved clinical outcomes continue to be demonstrated.⁴
- The U.S. Department of Health and Human Services (HHS) Social Determinants of Health (SDoH)^{5,6} examines broader factors affecting patient well-being—including where people live, learn, work, and access resources.
- The five core domains of SDoH are:
 - Economic Stability
 - Education Access & Quality
 - Health Care Access & Quality
 - Neighborhood & Built Environment
 - Social & Community Context
- Despite growing evidence of home infusion, a gap remains in applying SDoH in this setting, especially regarding patient-reported benefits and challenges in Ig and biologic therapies administered in the home.

OBJECTIVES

- This study aimed to evaluate patient perceptions of the benefits and challenges associated with home infusion therapy, across Ig and other biologic therapies, through the lens of SDoH. It focused on a diverse patient population, encompassing varying medical conditions, racial groups, education levels, and geographical locations.
- We sought to analyze a 30-question survey, guided by the HHS SDoH framework, that explored various patient perceptions related to home infusion therapy. This approach aimed to provide a deeper understanding of how SDoH affects home infusion therapy beyond just medical benefit and for the following areas of interest: Access and Awareness to Care, Financial Responsibility, Site of Care Safety, Cultural Awareness, General Mood, and Perception Change of Home Infusion Therapy.

METHODS

- A 10% random sample of over 3,000 home infusion patients (n=300) was performed, with a nurse-administered survey completed by 268 home infusion patients between 04/01/2024 and 07/31/2024.
- Patients were managed by company-employed nurses for infusion or self-administered SC therapy training and those managed by nursing agency were excluded.
- The survey assessed financial and time impact, accessibility and awareness, mood improvement, cultural understanding, home safety, and an overall perception change of home infusion therapy.
- Survey responses were gathered using "Yes or No" options or a scaled range (1-10) with defined response anchors. Statistical analysis was performed to summarize key findings and explore any subgroup differences.

RESULTS

Table 1. Patient Demographics

Variable	Home Infusion Therapy Patients (n =268)
Sex	
Female	167 (62%)
Male	101 (38%)
Age	
Median (Range)	64 (2-92)
2-11	3 (1.1%)
12-21	5 (1.9%)
22-31	11 (4.1%)
32-41	32 (11.9%)
42-51	27 (10.1%)
52-61	48 (17.9%)
62-71	67 (25%)
72-81	64 (23.9%)
82-92	11 (4.1%)
Race	
African American	24 (9%)
American Indian/Alaska Native	6 (2.2%)
Asian	3 (1.1%)
Hispanic/Latino	9 (3.4%)
Unreported	7 (2.6%)
White	218 (81.3%)
White and American Indian/Alaska Native	1 (0.4%)
Geographic Location	
Midwest	25 (9.3%)
Northeast	2 (0.7%)
South	187 (69.8%)
West	54 (20.2%)
Highest Level of Education	
Bachelor's Degree	67 (25%)
Doctoral Degree	11 (4.1%)
High School/GED	60 (22.4%)
Master's Degree	25 (9.3%)
Some College	92 (34.3%)
Some High School or Below	13 (4.9%)
Days on Service at Time of Survey (Months)*	
Median (Range)	10 (0-99)
0-10	133 (49.6%)
10-20	70 (26.1%)
20-30	31 (11.6%)
30-40	17 (6.3%)
40-50	6 (2.2%)
50-60	5 (1.9%)
60-70	1 (0.4%)
80-90	2 (0.8%)
90-100	3 (1.1%)

Figure 1. Financial and Time Impact

Variable	Median (IQR)	% of Top Response Anchor
Did Patients Feel the Option of Home Infusion Therapy Provided a Time and Cost Savings? 1 = Not At All 10 = Yes A Lot	10 (10,10)	85
Did Patients Feel the Option of Home Infusion Therapy Helped Balance Work Schedule? 1 = Not At All 10 = Yes A Lot	10 (9,10)	73
Did Patients Feel Their Job, or Lack of One, Made it Difficult for Home Infusion Therapy? 1 = Not At All 10 = Yes A Lot	2 (1,6)	12

Table 2. Patient Diagnoses and Therapy Types

Variable	Home Infusion Therapy Patients (n =268)
Diagnosis	
Acute Transverse Myelitis	2 (0.7%)
Ataxia	1 (0.4%)
Autonomic Neuropathy	1 (0.4%)
Chronic Inflammatory Demyelinating Polyneuropathy	68 (25.4%)
Collagenous Colitis	1 (0.4%)
Crohn's Disease	25 (9.3%)
Demyelinating Disease of Central Nervous System, Unspecified	3 (1.1%)
Dermatomyositis	2 (0.7%)
Dermatopolymyositis	4 (1.5%)
Encephalitis	7 (2.6%)
Guillain-Barré Syndrome	2 (0.7%)
Hereditary and Idiopathic Neuropathy	1 (0.4%)
Idiopathic Progressive Neuropathy	3 (1.1%)
Inclusion Body Myositis	1 (0.4%)
Juvenile Dermatomyositis	1 (0.4%)
Lambert-Eaton Myasthenic Syndrome	1 (0.4%)
Multifocal Motor Neuropathy	5 (1.9%)
Multiple Sclerosis	7 (2.6%)
Myasthenia Gravis	44 (16.4%)
Myositis	1 (0.4%)
Neuromyelitis Optica Spectrum Disorder	1 (0.4%)
Other Symptoms and Signs Involving the Musculoskeletal System	1 (0.4%)
Polymyositis	8 (3.0%)
Polyneuropathy	6 (2.2%)
Primary and Secondary Immunodeficiency	50 (18.7%)
Rheumatoid Arthritis	1 (0.4%)
Sjögren's Syndrome	1 (0.4%)
Stiff-Person Syndrome	5 (1.9%)
Ulcerative Colitis	15 (5.6%)
Current Therapy	
IVIg	197 (73.5%)
SCIG	15 (5.6%)
Integrin Receptor Antagonist	13 (4.9%)
Interleukin-12 and -23 Antagonist	13 (4.9%)
Tumor Necrosis Factor (TNF) Blocker	12 (4.5%)
CD-20 Directed Cytolytic Antibody	6 (2.2%)
Complement Inhibitor	5 (1.9%)
Interleukin-23 Antagonist	4 (1.5%)
Neonatal Fc Receptor Blocker	3 (1.1%)

Figure 2a. Home Safety

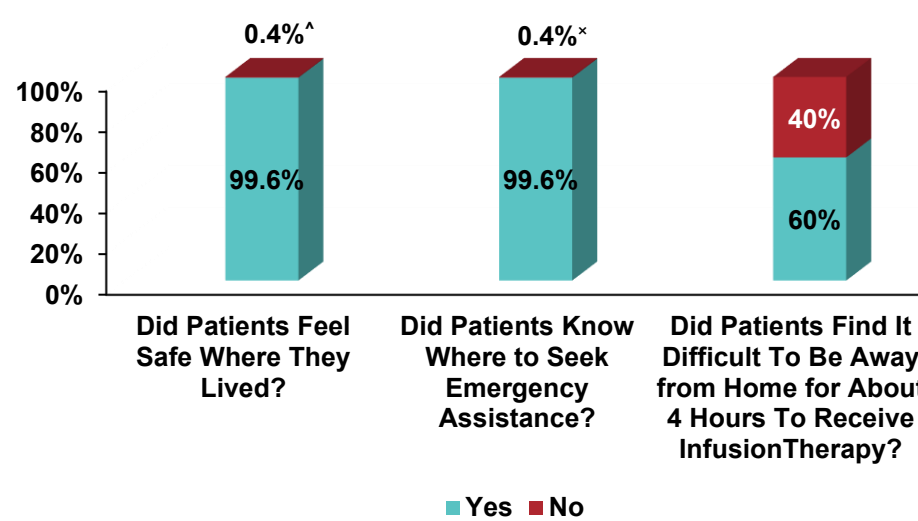


Figure 3a. Accessibility and Awareness

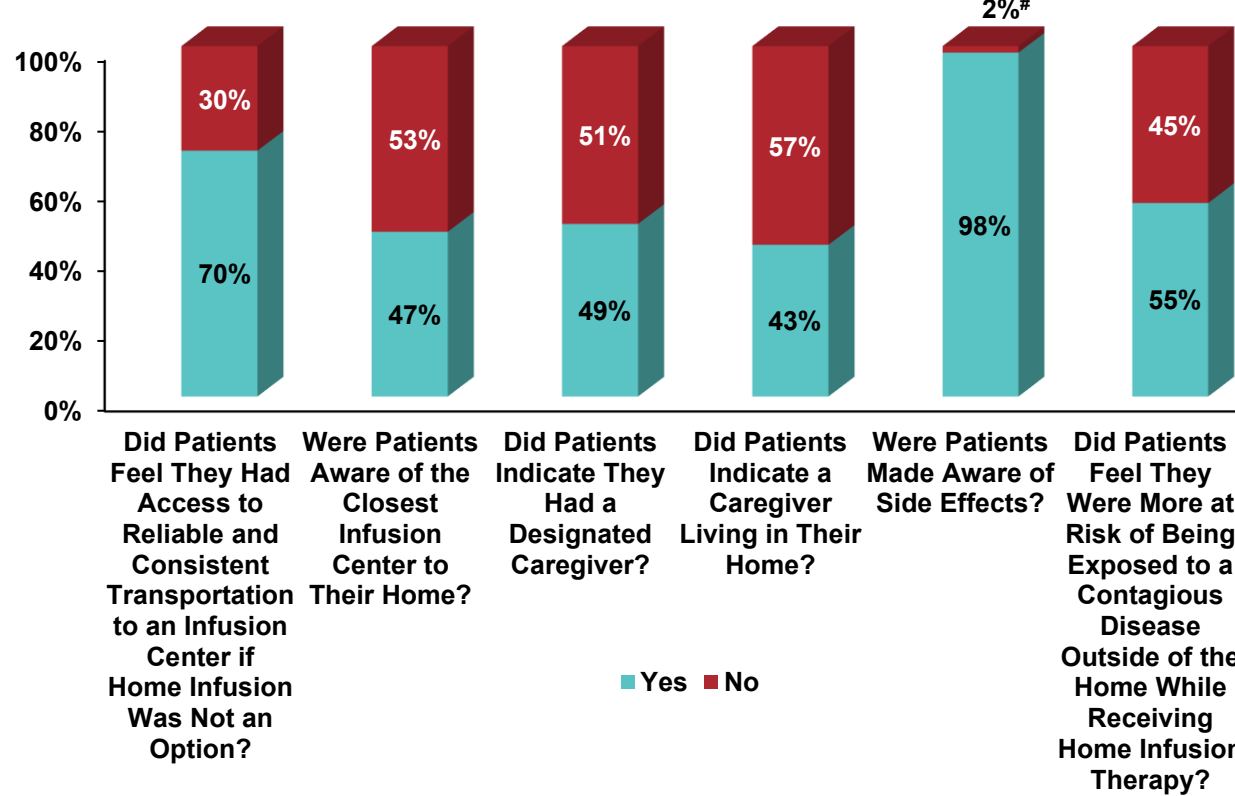


Figure 3b. Accessibility and Awareness

Variable	Median (IQR)	% of Top Response Anchor
How Difficult Did Patients Feel it Took to Pay for a Ride/Gas to Get IVIg Infusion Therapy Outside of Their Home? 1 = Not Hard At All 10 = Very Hard	4 (1,8)	19
How Sure Did Patients Feel Their Home Was a Safe and Clean Environment to Receive Infusion Therapy? 1 = Not Sure 10 = Very Sure	10 (10,10)	83
How Sure Did Patients Feel the Hospital/Infusion Center Was a Safe and Clean Place to Get Their Infusion Therapy? 1 = Not Sure 10 = Very Sure	5 (1,10)	26
How Sure Did Patients Feel if Something Went Wrong with Their Home Infusion Therapy, They Could Get Help? 1 = Not Sure 10 = Very Sure	10 (10,10)	79
How Easy Did Patients Feel it Was to Receive Home Infusion Therapy When They Needed it? 1 = Very Hard 10 = Very Easy	10 (10,10)	87
Did Patients Feel They Had Trouble Getting Their Infusion Therapy/Supplies When Needed in the Home Setting? 1 = Always 10 = Never	10 (10,10)	78
Did Patients Ever Have Trouble Getting Their Infusion Therapy/Supplies in the Hospital/ Infusion Center? 1 = Not At All 10 = Yes A Lot	1 (1,5)	8

Figure 2b. Home Safety

Variable	Median (IQR)	% of Top Response Anchor
How Easy Did Patients Feel it Was To Get Fresh Fruits and Vegetables? 1 = Not Hard At All 10 = Very Hard	1 (1,5)	7

Figure 4a. Cultural Understanding

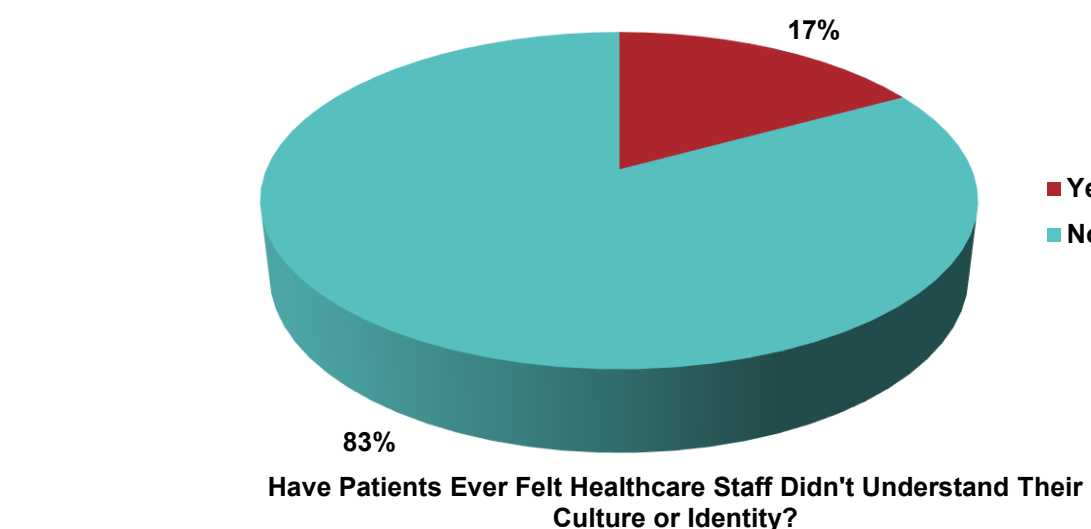


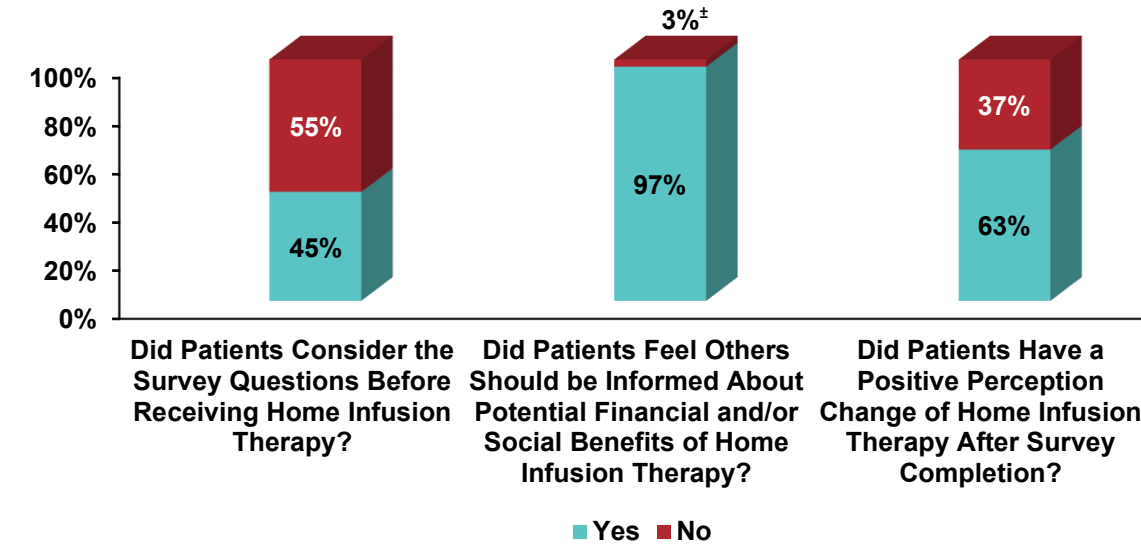
Figure 4b. Cultural Understanding

Variable	Median (IQR)	% of Top Response Anchor
How Important Did Patients Feel it is for Healthcare Providers to Understand Their Culture? 1 = Not Important 10 = Very Important	6 (1,10)	41
Did Patients Feel Home Infusion Helps with the Understanding of a Patient's Culture? 1 = Not Really 10 = Very Much So	10 (5,10)	57

Figure 5. Mood Improvement

Variable	Median (IQR)	% of Top Response Anchor
How Did Patients Report Feeling Most of the Time? 1 = Very Sad or Worried 10 = Very Happy and Safe	8 (7,10)	38
Did Patients who Received Home Infusion Therapy Improve Their Mood Compared to Traveling to a Hospital/Infusion Center for Treatment? 1 = Not Really 10 = Very Much So	10 (10,10)	81

Figures 6a. Perception Change of Home Infusion Therapy



Figures 6b. Perception Change of Home Infusion Therapy

Variable	Median (IQR)	% of Top Response Anchor
If a Positive Perception, Ranking of Change 1 = Slightly Changed 10 = Greatly Changed	9 (6,10)	45

CONCLUSION

This hybrid SDOH focused study is the first to explore home infusion perceptions across diverse demographic groups and disease states.

Patients reported overwhelmingly positive experiences of home infusion therapy, citing significant benefits in time and cost savings, work-life balance, access, cultural awareness, mental health, and safety.

Key Takeaways:

- Time and Cost Savings:
 - Recognized as a cost-effective, time-saving alternative to facility-based care
 - Notable improvements in work-life balance when treatment occurs at home
 - Employment-related barriers reported minimal for most, though a small subset faces challenges
- Access and Awareness
 - High reporting of reliable transportation to infusion centers if home infusion was not available, but logistical and cost barriers persist and only about half know the location of the nearest infusion center, indicating awareness gaps
 - Designated and in-home caregiver support is inconsistent, signaling inconsistencies in support availability
 - Home infusion supply delivery proves more dependable than facility-based logistics
 - Confidence in home safety and cleanliness far exceeds that in hospitals or infusion centers
 - Most patients felt highly confident they could receive help if complications occurred during home therapy
 - Facility-based care raises greater concerns about disease exposure and supply delays
- Emotional Well-Being
 - Home infusion therapy substantially enhances emotional well-being and overall satisfaction in infusion treatment
- Cultural Understanding
 - Cultural competence is crucial to patient experience and perceived quality of care and while most feel their cultural identity is respected, disparities exist
 - Home-based care offers opportunities to deepen cultural connection with providers and targeted interventions can further strengthen culturally responsive practices
- Home Safety
 - Findings affirm home environments as well-suited for safe, patient-centered disease management, supported by overwhelming sense of security and confidence in accessing emergency support should complications arise
 - Travel burden (e.g., travel time, treatment duration, and logistical hurdles) disproportionately affects certain patients, underscoring access gaps
 - Fresh-food access is generally manageable for most, though localized challenges persist
- Changes in Patient Perceptions of Home Infusion Therapy
 - Pre-treatment awareness of home infusion benefits varies (e.g., financial and social aspects), revealing education opportunities
 - Nearly all patients advocate informing others about home infusion advantages
 - Survey participation itself drives meaningful, positive shifts in patient perception of home infusion therapy
 - Emphasizing non-clinical benefits alongside clinical outcomes boosts treatment endorsement

Home infusion's financial, access, cultural, emotional, and safety benefits can enhance patient quality of life and reduce caregiver burden. Widespread positive perceptions suggest that expanding home infusion access may improve clinical outcomes and alleviate pressure on the healthcare system.

Due to the findings of this study, future research should focus on the following:

- Compare infusion treatment differences in patient perceptions, satisfaction, and long-term clinical outcomes between home infusion providers and traditional settings across broader populations
- Conduct cost-benefit analyses for both patient and healthcare systems (e.g., indirect costs like travel, missed work, caregiver burden, and mental health)
- Develop and evaluate strategies to mitigate home infusion access barriers, to home infusion therapy, specifically in underserved areas and rural populations
- Explore how education and awareness campaigns influence patient understanding, comfort, and adoption of home infusion therapy

By positioning home infusion therapy as a patient-centered and adaptable care model, we can improve the evolving preferences and long-term needs in modern healthcare.

- National Association of Specialty Pharmacy. NASP Definitions of Specialty Pharmacy and Specialty Medications. Accessed June 2025. https://naspnet.org/wp-content/uploads/2017/02/NASP-Definitions-final-2_16.pdf.
- Haines D. Cost Savings: Home Versus Inpatient Infusion Therapy. A Review of the Literature. *Infusion Journal*. November 2023;2(3):3-7.
- Polinski, JM, Kowal, MK, Gagnon M, et al. Home Infusion: Safety, Clinically Effective, Patient Preferred, and Cost Saving. *Healthcare*. 2017;5(1-2):68-80.

- American Society of Health-System Pharmacists. ASHP Guidelines on the Management of Home Infusion Services. *American Journal of Health-System Pharmacy*. 2022;79(5):380-388.
- Centers for Disease Control and Prevention. Social Determinants of Health. Public Health Gateway. Accessed June 2025. <https://www.cdc.gov/public-health-gateway/php/about/social-determinants-of-health.html>.
- U.S. Department of Health and Human Services. Healthy People 2030 – Social Determinants of Health. Accessed June 2025. <https://health.gov/healthypeople/objectives-and-data/social-determinants-health>.